

Department of Electrical and Electronic Technology
Faculty of Technology
Rajarata University of Sri Lanka
Academic Year 2023/2024
Semester II

CMT 1307 - Mathematics for Technology I
Assignment 06

1. State which of the following are vectors and which are scalars:

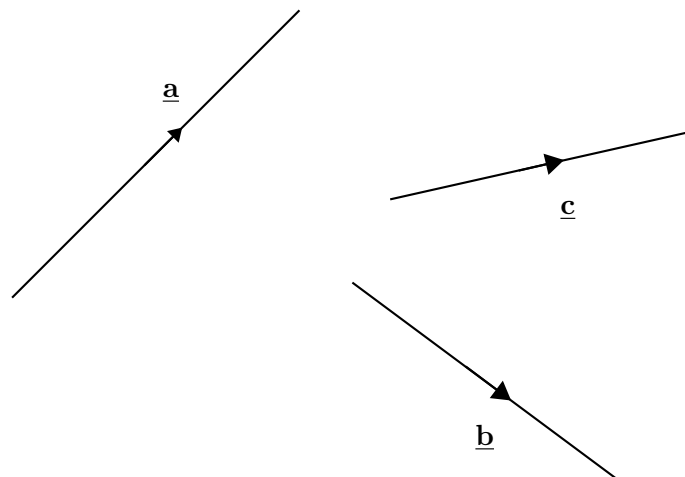
- | | |
|---------------|--------------------|
| (i) volume | (vi) distance |
| (ii) speed | (vii) displacement |
| (iii) force | (viii) temperature |
| (iv) velocity | (ix) work |
| (v) energy | (x) acceleration |

2. Let ABCD be a quadrilateral. In each of the following, find a single vector equal to the given vector expression:

- (i) $\overrightarrow{AB} + \overrightarrow{BC}$
- (ii) $\overrightarrow{CD} + \overrightarrow{DA}$
- (iii) $\overrightarrow{BA} + \overrightarrow{AD} + \overrightarrow{DC}$
- (iv) $\overrightarrow{AC} + \overrightarrow{DA}$
- (v) $\overrightarrow{AD} - \overrightarrow{CD}$

3. Let ABCD be a quadrilateral and let P, Q, R and S be the mid-points of the sides AB, BC, CD and DA. Using vector methods, show that PQRS is a parallelogram.

4. Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be the vectors shown in the figure.



construct

- (i) $\vec{a} + \vec{b} - \vec{c}$
 - (ii) $\vec{a} - 2\vec{b} + \vec{c}$
 - (iii) $\vec{c} - \frac{1}{2}(3\vec{a} - \vec{b})$
5. Using vector methods, prove that the line joining the mid-points of two sides of a triangle is parallel to the third side and has $\frac{1}{2}$ of its length.
6. Let ABC be a triangle and let O be a given point. Let P, Q and R be the mid-points of the sides AB, BC and CA, respectively. Prove that

$$\vec{OA} + \vec{OB} + \vec{OC} = \vec{OP} + \vec{OQ} + \vec{OR}.$$

7. Prove that if $\vec{a}$ and $\vec{b}$ are non-zero and non-parallel vectors satisfying $\lambda\vec{a} + \mu\vec{b} = \vec{0}$, then $\lambda = 0$ and $\mu = 0$.
8. Let OACB be a parallelogram and let D be the mid-point of AC. Let $\vec{OA} = \vec{a}$ and $\vec{OB} = \vec{b}$. Show that $\vec{OD} = \vec{a} + \frac{1}{2}\vec{b}$ and $\vec{AB} = \vec{b} - \vec{a}$. Let E be the point of intersection of OD and AB. Explain why $\vec{OE} = k_1\vec{OD}$ and $\vec{AE} = k_2\vec{AB}$, where k_1 and k_2 are scalars. Using these and the triangle law of addition to $\triangle OAE$, find the values of k_1 and k_2 .
9. Let OABC be a tetrahedron, and let $\vec{OA} = \vec{a}$, $\vec{OB} = \vec{b}$ and $\vec{OC} = \vec{c}$. Find $\vec{AB}$, $\vec{BC}$ and $\vec{AC}$ in terms of $\vec{a}$, $\vec{b}$ and $\vec{c}$. Let D be the mid-point of OA and E be the point on BC such that BE:EC=2:1. Find $\vec{DE}$ in terms of $\vec{a}$, $\vec{b}$ and $\vec{c}$.

10. Relative to a fixed origin O, the position vectors of the points A, B and C are given to be $\vec{i} + \vec{j} + 2\vec{k}$, $2\vec{i} - \vec{j} + 3\vec{k}$ and $2\vec{j} + \vec{k}$.
- (i) Find $\overrightarrow{AB}$, $\overrightarrow{AC}$, $|\overrightarrow{AB}|$ and $|\overrightarrow{AC}|$.
- (ii) Find the angle $\widehat{BAC}$.
11. Let $A \equiv (1, 2, -1)$ and $B \equiv (2, -1, 3)$.
- Write down $\overrightarrow{OA}$ and $\overrightarrow{OB}$ in terms of $\vec{i}$, $\vec{j}$ and $\vec{k}$, where O is the origin.
- Find $\overrightarrow{AB}$ and the unit vector in the direction of $\overrightarrow{AB}$.
- Find $\overrightarrow{OA} \cdot \overrightarrow{OB}$ and hence, find the angle $\widehat{AOB}$.
- Also find the angle $\widehat{OAB}$.
12. (a) Let $\vec{a} = 2\vec{i} - 3\vec{j} + \vec{k}$ and $\vec{b} = \vec{i} - 2\vec{j} - 8\vec{k}$. Show that $\vec{a}$ and $\vec{b}$ are perpendicular.
- (b) Let $2\vec{i} + \vec{j}$ and $\vec{i} - 3\vec{j} + \vec{k}$ be the position vectors of two points A and B respectively, with respect to a fixed origin O. Let C be on the straight line OB such that $\widehat{OCA} = \frac{\pi}{2}$. Find $\overrightarrow{OC}$ in terms of $\vec{i}$, $\vec{j}$ and $\vec{k}$.

Answer question 12 and submit your answer scripts on or before 24th September 2026 at 4.30 pm.